

# From Sparse Selection to Physical Savings: A Runtime Contract for Persistent Native Memory

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## Abstract

Logical context sparsity is only an opportunity bound. A runtime may select a small history while rebuilding K/V, copying selected tensors, changing source positions, transferring bytes, constructing K/V-sized temporaries, or inducing additional model calls. We define a vector-valued realization contract for persistent native memory that qualifies semantic same-subset correctness, original-position consumption, resident-state reuse, physical movement, lifecycle, and end-to-end workload economics independently.

The corrected implementations realize disjoint resident K/V through native views or scheduler-owned pages in bounded HF, MLX, SGLang, llama.cpp, vLLM-Metal, and vLLM-CUDA scopes. Matched-consumer probes report zero retained- history re-encoding and zero selected-K/V packing while separately exposing host/device transfer, offload, and bounded attention scratch. The correction ledger shows why these fields cannot be collapsed into “zero-copy”: earlier paths reconstructed caches, used compact rather than original positions, performed hidden casts or copies, and created K/V-sized temporaries.

With frozen RAG selections on Qwen3-4B MLX, cold native execution is 3.2% slower than selected text, but warm repeated selection reduces mean TTFT from 2.640 to 0.166 seconds and total latency from 3.031 to 0.638 seconds. Ordinary exact prefix-cache reuse remains a required baseline; PRA’s distinct advantage is changing or detached addressable evidence. On one fresh locked coding task, MLX and SGLang-MLX PRA-100 reproduce all six plain-engine actions and the exact patch without re-encoding selected history or copying selected K/V. Their task-aware 90% treatments also preserve the solve and six-call count, but realize only 2.19% cumulative physical-input saving on the short trajectory; SGLang additionally exposes a 17.5 MB suffix-graft copy and 1.61 GB peak consumer temporary. A separate frozen plan omits about 40% of resident history, but its receipt-bearing transfers remain hybrid transformed-context evidence rather than strict resident-subset reuse.

The resulting measurement contract establishes two boundaries: logical selection does not imply physical saving, and physical omission does not imply end-to-end workload saving.

## 1 Introduction

PRA changes what a model must keep logically available. A long-lived resource may remain addressable through a compact routing representation while only a small selected part of its native K/V participates in one attention operation. That distinction is algorithmically useful, but it leaves a systems question unanswered. A selected set can be small while the runtime still stores every detail tensor in HBM, launches many tiny gathers, copies the same bytes repeatedly, constructs temporary concatenations, or serializes requests around Python control flow. In that case sparsity exists in a trace but not in the resource bill.

This paper asks how to turn PRA into a portable inference primitive before a serving engine becomes semantically aware of PRA memory. The immediate task is not to design a new scheduler.

It is to define an execution boundary that can be measured, optimized, and later replaced without changing the meaning of a selected reference.

The runtime composes previously defined model, resource, and session interfaces, but this paper evaluates only the native realization boundary. Policy, transport, storage, and tool surfaces remain separate contracts; their product details are documented in the appendix and companion papers rather than treated as additional systems claims here.

The paper is organized around four independent gates:

- **Semantic gate.** The engine must consume the same selected state as an identical-subset reference from the same resident source.
- **Position gate.** Sparse records retain original source positions; compact selected width is never substituted for the source position base.
- **Physical mechanism gate.** Re-encoding, selected-K/V copy, host–device transfer, consumer temporaries, ownership, and lifecycle are reported independently.
- **Workload gate.** TTFT, total latency, calls, completion work, and task outcome determine whether physical omission becomes an economic saving.

The companion Paper 8.5 owns engine-independent agent-memory discovery, including record exclusion, causal policy, oracle add-back, and logical task efficiency. Paper 4.5 imports a frozen, hash-bound **AgentMemoryPlan**; it does not tune or promote the selector. Full history and 100% retention remain correctness controls before any reduced plan is interpreted.

More precisely, Paper 8.5 decides which logical records or observation spans remain visible. Paper 4.5 consumes a frozen, hashable **AgentMemoryPlan** and tests whether an engine realizes that identical plan from resident K/V with original positions, no selected-history re-encoding or physical copy, safe lifecycle, and separately reported temporary and runtime costs. Engine outcomes do not retune the selector.

The unified runtime, gateway, authenticated resource lifecycle, typed tool and result surfaces, and engine-specific adapters are the implementation needed to test those claims. They are not additional evidence categories. In particular, this paper treats correction of an overstated result as evidence about the measurement contract: cache identity, residence, source position, physical copy, and realized retention must be observed independently.

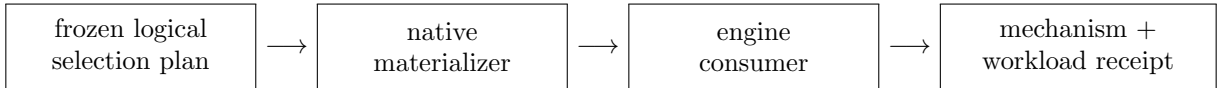


Figure 1: Paper boundary. Policy chooses stable records and spans; the runtime must realize that frozen plan and expose the physical work actually performed.

## 2 Accounting Model

For  $L$  decoder layers,  $T$  source tokens,  $H_{kv}$  physical K/V heads, head width  $D$ , and  $b$  bytes per scalar, complete detail storage is

$$B_{\text{detail}} = 2|L_{KV}|H_{kv}TD b.$$

If request-time selection admits  $S$  tokens to  $|L_C|$  consuming layers, the active payload opportunity is

$$B_{\text{active}} = 2|L_C|H_{kv}SD b.$$

The ratio  $B_{\text{active}}/B_{\text{detail}}$  is not peak HBM or runtime. Weights, local K/V, allocator state, routing indices, transfer staging, score buffers, and temporary packing remain outside it.

| Quantity                     | Interpretation                                                                      |
|------------------------------|-------------------------------------------------------------------------------------|
| Full/selected logical tokens | Policy opportunity before engine realization.                                       |
| Visible prompt tokens        | Text actually submitted through the ordinary prompt path.                           |
| Resident/selected source K/V | Canonical stored extent and the intervals read by this request.                     |
| Original-history re-encoding | Old selected tokens evaluated through the model again.                              |
| Synthetic-history encoding   | Post-cache summaries, tombstones, or receipts evaluated as new tokens.              |
| Partial-old-history encoding | Previously seen but uncached tails evaluated again, for example an incomplete page. |
| Selected-K/V copy and H2D    | Explicit packing/copy and residency transfer, reported separately.                  |
| Consumer temporary bytes     | Peak request-local score, reduction, concatenation, and allocator state.            |
| Workload                     | Requests, TTFT, total latency, completion tokens, peak memory, and task outcome.    |

Table 1: Independent accounting quantities. No single “reuse” or “zero-copy” label substitutes for these fields.

Evidence is reported in four classes. A *semantic gate* verifies that the selected state produces the intended same-subset result. A *mechanism gate* observes positions, historical evaluation, copying, and ownership. A *physical-efficiency gate* measures resident bytes, transfer, temporaries, and peak memory. A *workload gate* measures requests, TTFT, total latency, completion work, and task outcome. An engine can pass the first two and still fail the latter two; the vector must not be collapsed into “PRA supported.”

The accounting receipt is request-scoped and stage-labelled. Bytes copied to bootstrap a canonical source are different from bytes copied for every selection. Similarly, one-time ingest encoding is different from re-encoding already resident history. Peak allocation is reported separately from cumulative traffic, because a small reusable scratch buffer and a newly packed K/V tensor can have similar peaks but radically different repeated-request costs.

### 3 Native Realization Contract

The core contract is `live-kv-original-position-lifecycle-v1`. A request identifies the exact source cache, runtime/model/tokenizer revision, stable selected records and spans, original logical positions, session and tenant scope, and a plan digest. A same-subset comparator consumes identical K/V values at identical positions. Ownership remains with the canonical source while requests borrow intervals; finish, cancellation, error, eviction, offload/restore, termination, and stale generation rejection are independently tested.

Qualification is vector-valued:

$$G = (S, P, C, L, E),$$

where  $S$  is same-state semantic correctness,  $P$  is original-position sparse correctness,  $C$  is copy/re-encoding behavior,  $L$  is lifecycle correctness, and  $E$  is end-to-end workload qualification. Passing one coordinate does not promote another.

After one source bootstrap, strict resident-subset execution permits only genuinely new history delta, the current suffix, and generated tokens. It requires

$$R_{\text{orig}} = R_{\text{synthetic}} = R_{\text{partial}} = 0, \quad C_{\text{selectedKV}} = 0.$$

Ingest-time compaction may create reusable K/V before source admission. Replacing already resident records with new receipt text is instead a hybrid transformed-context treatment and cannot qualify this invariant.

### 3.1 Position and composition semantics

The plan carries source/original positions, while an implementation may also have chunk-local and request-packed coordinates. These are not interchangeable. For rotary models, a cached key associated with logical position  $p$  contains the corresponding rotation (schematically  $K_{\text{post}}(p) = R_p K_{\text{pre}}$ ). Relabelling an already rotated key or compacting disjoint records into positions  $0, \dots, S - 1$  changes the attention state. The same-subset comparator therefore uses the identical selected K/V and original positions; it does not compare sparse original-position execution with a newly packed text prompt and call any difference an implementation bug.

Composition is separately declared. Independently encoding chunks at local positions can be a valid alternative representation, but it is a different treatment from globally encoding the concatenated document. The engine must name which representation it consumes and qualify quality under that representation rather than infer equivalence from token identity.

### 3.2 Ownership and lifecycle

The canonical source owns resident K/V; requests borrow immutable intervals or pages. A lifecycle gate covers concurrent session isolation, cancellation, normal termination, stale-fork rejection, eviction, offload/restore, and final cleanup. Releasing one borrower must not free state still referenced by another, while eviction must either wait for borrowers or move through a generation-checked handoff. These tests are mechanism evidence, not throughput benchmarks, but they are required before an aliasing implementation can be called reusable.

Paper 8.5 supplies the following frozen handoff:

```
AgentMemoryPlan
  selected stable record IDs and spans
  original logical positions
  materialization/compaction stage
  source and plan digests
```

The engine realizes this plan; it does not infer task semantics or retune the policy.

## 4 Runtime Architecture

The contract is implemented by a narrow set of components. The application or Paper 8.5 policy emits a logical plan; a resource registry resolves stable identities to a canonical source generation; a materializer validates spans and positions; an engine adapter creates non-owning views or scheduler aliases; a lifecycle manager owns eviction and restoration; and telemetry records the work performed. Tool schemas, generic gateway UX, and policy tuning are not part of this core path.

| Component         | Authority                                    | Required receipt                                         |
|-------------------|----------------------------------------------|----------------------------------------------------------|
| Selection plan    | stable record IDs, spans, positions, digest  | selected logical extent and materialization stage        |
| Resource registry | tenant/session/model/source generation       | exact source-cache identity and residency state          |
| Materializer      | interval validation and consumer descriptor  | rejected spans, old/synthetic/partial encoding counts    |
| Engine adapter    | attention consumer and native alias geometry | K/V copy, transfer, temporary, and position mode         |
| Lifecycle manager | borrow, release, eviction, restore, cleanup  | ownership generation and terminal-state outcome          |
| Workload harness  | prefix/full controls and task execution      | TTFT, latency, calls, completion work, peak, and quality |

Table 2: Minimal architecture supporting the native realization contract.

A plan digest binds policy output to runtime evidence. If an adapter changes the selected intervals, compacts positions, inserts receipt text, or falls back to prompt replay, it must emit a distinct realization digest and accounting class. This makes silent degradation observable and permits the same frozen logical treatment to be compared across engines.

The ordinary prefix cache remains available beside addressable resources. Local request deltas should continue through the engine’s normal incremental path; persistent source records remain separately addressable. PRA is useful when the next request needs a disjoint or changing subset that cannot be represented as one unchanged prefix. It should not replace an exact-prefix hit with a more expensive sparse path.

## 5 Corrected Engine Matrix

| Engine     | Model/backend scope            | <i>S</i> | <i>P</i> | <i>C</i> | <i>L</i> | <i>E</i> | Primary remaining limitation                                                                            |
|------------|--------------------------------|----------|----------|----------|----------|----------|---------------------------------------------------------------------------------------------------------|
| HF         | Qwen/CUDA, batch 1             | pass     | pass     | pass     | pass     | open     | Autonomous frozen-plan workload not rerun after the final contract.                                     |
| MLX        | Direct MLX, batch 1            | pass     | pass     | pass     | pass     | 1 task   | Multi-task and cross-model economics remain open; selected-copy is zero but other K/V movement remains. |
| SGLang-MLX | Fused disjoint MLX consumer    | pass     | pass     | pass     | pass     | 1 task   | Multi-task economics remain open; reduced execution has large consumer scratch.                         |
| llama.cpp  | Resident slot/sequence ranges  | pass     | pass     | pass     | pass     | 1 task   | Endpoint temporary allocation remains incomplete; current transfer is hybrid.                           |
| vLLM-Metal | Bounded page geometry          | pass     | pass     | pass     | 24/24    | open     | No frozen autonomous reduced-plan workload row.                                                         |
| vLLM CUDA  | V1, complete homogeneous pages | pass     | pass     | bounded  | pass     | open     | Frozen-plan bridge still needs an autonomous no-copy workload closure.                                  |

Table 3: Vector-valued qualification  $G = (S, P, C, L, E)$ : same-state semantics, original positions, copy/re-encoding, lifecycle, and autonomous workload. “Pass” is limited to the declared scope; “diag.” and “open” are not inferred from mechanism success.

The primary correction ledger is consolidated in Appendix A. Earlier rows that reconstructed caches, used compact selected positions, performed hidden casts/copies, or created K/V-sized temporaries remain debugging evidence only.

## 6 Results

### 6.1 Warm repeated selection produces a real native-memory benefit

The cleanest realization experiment freezes 50 Qwen3-4B MLX RAG selections at 2K context. Selected text and native memory match every output and quality metric. Cold native execution costs 3.121 versus 3.024 seconds, a 3.2% regression from native encoding. Once the identical selection is resident, mean TTFT falls from 2.640 to 0.166 seconds and total latency from 3.031 to 0.638 seconds; throughput rises from 10.56 to 50.21 tokens/s. At 4K, cold native is 3.3% slower while warm total latency is 0.799 versus 6.121 seconds. Qwen3-8B repeats the pattern: 1.8% cold regression, then an 82.1% warm total-latency reduction. These are repeated-selection results on one M4 host, not general serving speedups.

| Model / context         | Cold selected | Cold native | Warm selected | Warm native |
|-------------------------|---------------|-------------|---------------|-------------|
| Qwen3-4B / 2K total (s) | 3.024         | 3.121       | 3.031         | 0.638       |
| Qwen3-4B / 2K TTFT (s)  | —             | —           | 2.640         | 0.166       |
| Qwen3-4B / 4K total (s) | —             | +3.3%       | 6.121         | 0.799       |
| Qwen3-8B / 2K total     | —             | +1.8%       | baseline      | −82.1%      |

Table 4: Repeated-selection realization summary. Dashes indicate that the main claim uses the reported relative effect rather than reconstructing an unreported absolute cell.

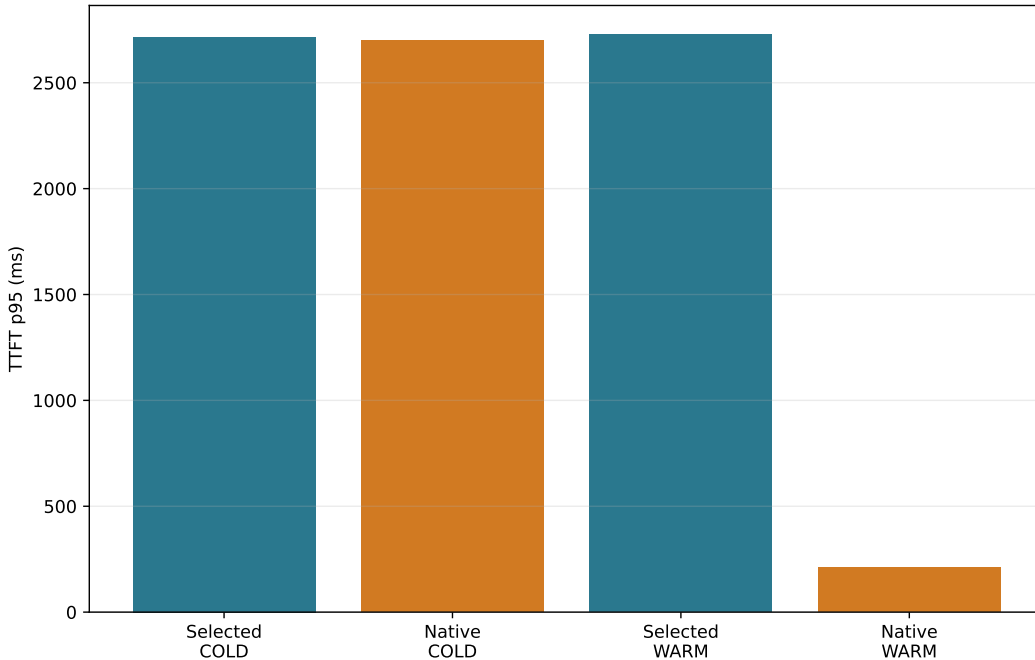


Figure 2: Qwen3-4B, 2K, 50-query mechanism cohort. Cold native realization has no material TTFT advantage; warm residence removes repeated selected-text prefill. Values are host- and cohort-specific.

### 6.2 Exact prefix caching is a necessary baseline

Warm native memory must not be compared only with fresh text prefill. The required ladder is fresh selected text, exact text-prefix hit, persistent native selection, and changing-selection partial native reuse. An autonomous llama.cpp control illustrates the strength of the baseline: at 100% retention,

ordinary live-prefix continuation matches cold-prefill behavior while reducing single-task wall time by  $27.85\times$ . This is a prefix-cache result, not a PRA speedup. PRA’s distinct systems opportunity begins when selected evidence is detached, disjoint, or changes across requests.

### 6.3 Complete retention must remain on the native dense path

The corrected HF dispatcher no longer sends PRA-100 through the disjoint online-softmax consumer. It shallow-forks only request-local cache descriptors while every key/value tensor continues to alias the pinned canonical source; request growth replaces private descriptors without mutating that source. Transformers therefore executes its ordinary dense attention and last-logit projection geometry. A real Qwen2.5-0.5B FP16 control with 2,713 resident source tokens is token exact and has maximum logit delta 0.0 against two independent ordinary-prefix executions and after lossless restore. It reports zero selected-history re-encoding, selected-K/V copy, interval packing, or consumer K/V temporary. This is a bounded correctness result, not an autonomous workload or latency claim. Earlier PRA-100 probes that forced a complete contiguous prefix through a sparse consumer remain useful numerical diagnostics but are not admissible full-retention implementations. The machine-readable receipt is `shared/results/paper4_5_runtime_productization/coding_agents/engine_gates/hf_qwen25_05b_full_native_dense_fix_20260919.json`.

### 6.4 Changing selection changes composition semantics

Persistent-corpus MLX reuses independently encoded 256-token chunks across 50 changing questions. It reuses 22.0% of chunk lookups and reduces cumulative request time by 12.8%, but official score changes from 0.440 to 0.380. The cause is representational: chunk-local post-RoPE positions differ from the single global position sequence of packed text. This is not cache corruption. It establishes

same selected evidence  $\not\Rightarrow$  same hidden-state semantics.

Original, chunk-local, and request-packed positions are therefore explicit contract fields rather than implementation details.

### 6.5 Fresh autonomous controls separate mechanism from policy

An earlier SGLang diagnostic sent identical tokens through different numerical consumer paths and diverged at request three. That row motivated the matched- consumer requirement but is superseded as engine qualification: complete retention must remain on the ordinary dense path. The resulting rule is

same token IDs  $\not\Rightarrow$  same numerical consumer path.

Comparisons must therefore identify both selected state and its consumer.

Fresh autonomous MLX and SGLang-MLX controls then hold the model revision, temperature-zero decoding, task, container, and grader fixed. On both engines, plain and PRA-100 resolve in the same six actions and submit the byte-identical patch; all six assistant actions match exactly, with zero selected-history re-encoding and zero selected-K/V copy. A matched whole-turn token tail on MLX fails after nine calls, while the structural progress/mutation/verification- floor policy resolves on both engines in six calls with the reference patch. It first diverges only in final-response wording and realizes 8,725 physical tokens from 8,920 logical tokens, a 2.19% saving.

The same matched-tail rule is not globally invalid. A later engine-independent Paper 8.5 transfer with OpenHands and Qwen3-Coder-30B preserves 3/3 official resolutions, reduces actions from 114 to 79, and saves 46.86% against paired FULL input (commit `ad1152b9`). The contrast establishes

agent/model/task dependence: a qualified logical policy can still fail a particular native-engine workload pair. Paper 4.5 imports only its whole-record selection. Paper 8.5’s extra slicing of oversized text observations is not a native-K/V saving unless record-internal original-position spans are consumed without re-encoding; the paired 46.86% also includes trajectory contraction and is not an engine-local K/V or latency number.

The engines nevertheless have different physical realization costs. Direct MLX reports 8.40 GB on its cumulative total-K/V-copy counter, despite zero selected-K/V copy, and a 1.25 GB peak consumer temporary for the reduced run. Fused SGLang-MLX reduces its total-K/V-copy counter to a 17.5 MB canonical- suffix graft, but its reduced-path consumer temporary peaks at 1.61 GB. These are one-task correctness and policy- sensitivity results, not useful workload-economics claims. The fresh SGLang evidence is archived under `shared/results/paper4_5_runtime_productization/coding_agents/engine_agent_gates/sglang_task01_qwen25coder14b_postfix_6e97f0c6/`.

## 6.6 Frozen agent-plan transfer separates omission from economics

Paper 8.5 supplies one Prompt Pinned plan on Scikit-learn-14496. The current transfers are hybrid because they create model-visible closure receipts after the source cache exists; they qualify original-K/V selection but not the strict resident-subset invariant.

| Engine     | PRA-100 | Reduced | Calls F/R | Logical/KV omission | Hist. re-enc.         | K/V copy   | Paired workload | Status                       |
|------------|---------|---------|-----------|---------------------|-----------------------|------------|-----------------|------------------------------|
| llama.cpp  | pass    | solve   | 8/8       | 40.23/40.23%        | 0 orig.; receipts n/a | 0 selected | calls preserved | hybrid; timing open          |
| MLX        | pass    | solve   | 9/12      | 39.45/40.04%        | 1,056 synth.          | 0 selected | +15.23% saving  | hybrid                       |
| SGLang-MLX | pass    | solve   | 7/15      | 39.45/37.21%        | 1,344 synth.          | 0 selected | −33.10% saving  | hybrid                       |
| HF CUDA    | mech.   | open    | —         | —                   | —                     | —          | open            | autonomous gate open         |
| vLLM CUDA  | mech.   | open    | —         | —                   | —                     | —          | open            | autonomous no-copy gate open |

Table 5: Frozen-plan workload economics. F/R denotes matched Full/reduced calls. Positive paired workload is saving; negative is regression. Receipt rows are hybrid transformed context and are not pooled with strict resident- subset execution.

These rows establish the second systems boundary:

$$\boxed{\text{physical omission} \not\Rightarrow \text{end-to-end workload saving.}}$$

Selection-only reruns with zero synthetic history are required before the plan can enter the strict native class.

## 7 Prefix Cache and PRA Are Complementary

Exact prefix caching is the correct default when the next request extends one unchanged token prefix. It preserves the engine’s native consumer path and can reuse all prefix K/V with minimal planning. PRA addresses a different shape: the request needs a changing, detached, or disjoint subset of already resident records. A runtime should choose between the two using observable request geometry, not route every request through sparse selection.

| Request shape                   | Preferred path              | Required baseline                                     | Main risk                                                |
|---------------------------------|-----------------------------|-------------------------------------------------------|----------------------------------------------------------|
| unchanged prefix plus suffix    | exact prefix cache          | live-prefix continuation                              | replacing a near-perfect hit with planning/copy overhead |
| same frozen detached evidence   | persistent native selection | warm selected-text and prefix hit where representable | residence cost and consumer temporaries                  |
| changing disjoint records       | partial native PRA          | fresh selected text plus matched consumer             | position/composition mismatch and churn                  |
| post-cache summary or receipt   | hybrid transformed context  | selected K/V plus charged synthetic encoding          | describing new text as cache reuse                       |
| uncached or partial source tail | mixed/fallback path         | explicit partial-history accounting                   | hiding old-token evaluation in a nominal hit             |

Table 6: Dispatch rule and comparator ladder by request geometry.

This division also clarifies the value proposition. PRA does not need to beat prefix caching on unchanged prefixes; it must outperform rebuilding or copying the selected state when ordinary prefix identity no longer applies. The telemetry receipt therefore records exact-prefix reused tokens and addressable resident intervals separately. A combined runtime can keep the local agent delta on the prefix path while borrowing older resource intervals through PRA, without re-encoding either class.

## 8 Qualification and Promotion Procedure

Qualification proceeds from semantics to workload; later measurements cannot repair an earlier failure. First, a deterministic bounded geometry compares native sparse execution against a same-subset reference with identical K/V and positions. Second, instrumented counters require zero original-history evaluation and separately expose synthetic or partial-history work, selected K/V copies, transfers, and consumer temporaries. Third, lifecycle tests cover concurrent borrowers and every terminal path. Fourth, cold, warm, and changing-selection cohorts measure physical and latency effects against both fresh prefill and exact prefix caching. Finally, an autonomous workload consumes a frozen policy plan without engine-specific retuning.

| Gate      | Required comparator                           | Promotion evidence                                                                    |
|-----------|-----------------------------------------------|---------------------------------------------------------------------------------------|
| Semantic  | identical selected K/V and original positions | bounded output/logit agreement under declared dtype and consumer path                 |
| Mechanism | instrumented resident source                  | zero old-history evaluation; copy, transfer, temporary, and lifecycle fields reported |
| Physical  | cold/warm/changing selection                  | resident and request-local bytes plus peak and cumulative traffic                     |
| Workload  | prefix cache and fresh prefill                | task outcome, requests, TTFT, latency, completion work, and paired cumulative cost    |

Table 7: Promotion ladder. A row is not inferred from success in a later row.

This ordering prevents three common category errors. Matching output alone does not prove that historical K/V was reused. Zero selected-K/V copy does not prove low temporary memory or low transfer. A fast request does not prove an end-to-end saving if the model takes more requests to solve the task.

## 9 Scope and Remaining Gates

The corrected adapters establish bounded mechanisms, not universal engine support. HF remains qualified only for declared Qwen/CUDA batch-one geometry; MLX and SGLang retain batch, topology, and head-dimension restrictions; llama.cpp still lacks complete endpoint temporary-allocation visibility; and vLLM CUDA remains limited to in-process V1, complete pages, and a homogeneous K/V group. Partial pages, hybrid groups, multiprocess scheduling, broader models, and production concurrency require separate gates.

The warm RAG result is a repeated-selection mechanism cohort on one Apple host. It demonstrates that correct residence can remove repeated prefill, but it does not establish portable serving speedups. The changing-selection experiment also shows why globally packed text cannot be treated as the universal semantic reference for independently encoded chunks.

The current agent transfer is one frozen task-plan identity and is hybrid because closure receipts are encoded after caching. Its value is diagnostic: similar original-K/V omission leads to preserved, reduced, or negative workload benefit across consumers. The strict next experiment removes synthetic history, holds the selected records fixed, and repeats the paired autonomous gate with task-clustered replication. HF and vLLM CUDA have bounded mechanism evidence but no admissible post-contract autonomous reduced-plan row; their workload coordinate  $E$  therefore remains explicitly open.

## 10 Conclusion

Sparse logical selection is an opportunity bound, not a systems result. Native reuse requires source identity, original positions, same-subset correctness, zero hidden historical evaluation, scoped K/V movement, bounded consumer state, and safe ownership. Correct persistent native memory can deliver large warm repeated-selection gains, while exact prefix caching remains a formidable baseline. Changing selections introduce composition semantics that ordinary prefix reuse does not solve. Finally, an agent can react to physically correct omission by taking more calls, erasing or reversing the saving. The runtime contract therefore ends at vector-valued evidence, not a single reuse flag.

## A Correction Ledger

The runtime contract was tightened after agent-history experiments exposed several ways to overstate native reuse. Table 8 records the corrections that invalidate earlier economic claims while preserving their value as debugging evidence. The complete chronology and raw tables remain in the technical-report edition.

| Earlier treatment     | Why it is invalid                                                   | Current requirement                                                          |
|-----------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------|
| Reconstructed cache   | Fresh prefill is not the same resident state.                       | Fork or alias the canonical live cache; report old-token evaluation.         |
| Compact position base | Disjoint records changed rotary/causal coordinates.                 | Preserve original source positions in the plan and consumer.                 |
| Hidden cast or pack   | Selected K/V was copied while reported as reuse.                    | Report selected-K/V copy, dtype conversion, and H2D separately.              |
| K/V-sized temporary   | Selection was view-based but attention concatenated resident K/V.   | Report consumer peak and cumulative temporaries independently.               |
| Retention ceiling     | Whole records rounded below a nominal percentage.                   | Use the authoritative tokenizer and fail closed below the declared floor.    |
| Mixed revisions       | Mechanism and workload rows combined different packages or patches. | Bind model, tokenizer, engine, PRA revision, source, and plan digests.       |
| Post-cache receipt    | New summary text was described as resident-subset reuse.            | Classify and charge transformed context separately from strict native reuse. |

Table 8: Authoritative correction ledger for publication claims.

## B Bounded Engine Scopes

Engine qualification is deliberately scoped rather than binary. HF covers Qwen2/Qwen3 full-attention models at batch one; MLX and SGLang-MLX cover their declared Metal topology; llama.cpp covers resident sequence/range attachment; and vLLM CUDA covers in-process V1 with complete homogeneous pages. These rows do not imply partial-page, hybrid-group, multiprocess, distributed, or production-concurrency support. A mechanism-qualified row retains  $E = \text{open}$  until a matched autonomous workload reports task outcome, calls, completion work, prefix-cache status, physical traffic, and repeated latency.

The frozen agent-policy handoff contains the policy ID, plan and source-prefix digests, selected stable record IDs, spans, original positions, and materialization class. Any runtime that changes one of those fields creates a new realization row. Selection-only resident subsets and receipt-bearing hybrid transformed context are never pooled.

## C Reproducibility Map

The implementation and evidence are rooted at the following repository paths:

- `src/prahf/hf_live_kv.py`, `src/pramlx/native.py`, `src/prasglang/`, and `src/pravllm/` implement the bounded native consumers and ownership contracts.
- `experiments/paper4_5_agent/` contains the matched FULL, PRA-100, frozen-plan, lifecycle, and autonomous workload runners.
- `docs/papers/shared/results/paper4_5_runtime_productization/agent_memory_plans/` contains hash-bound Paper 8.5 plans, engine receipts, and autonomous outcomes.
- `docs/papers/shared/results/paper4_5_runtime_productization/rag/` contains frozen-selection cold/warm realization evidence.

Every headline value in the publication edition maps to a machine-readable artifact under these roots. The technical-report edition retains the full implementation guide, gateway and storage studies, historical matrices, and superseded diagnostic tables.